EXP 1:

Design, Realization and Performance Analysis of Butterworth IIR Low-Pass and High-Pass Digital Filters Using MATLAB

Objective 1:

To design Butterworth IIR low-pass and high-pass digital filters for specified frequency requirements using MATLAB.

CODE:

```
clc;

clear;

close all;

N = 4;

Fs = 1000;

Fc = 100;

Wn = Fc/(Fs/2);

[b1,a1] = butter(N,Wn,'low');

[b2,a2] = butter(N,Wn,'high');

figure;

subplot(2,1,1);

freqz(b1,a1,1024,Fs);

title('Butterworth IIR Low-Pass Filter');

subplot(2,1,2);

freqz(b2,a2,1024,Fs);

title('Butterworth IIR High-Pass Filter');

disp('Low-Pass Filter Coefficients:');

disp(b1);

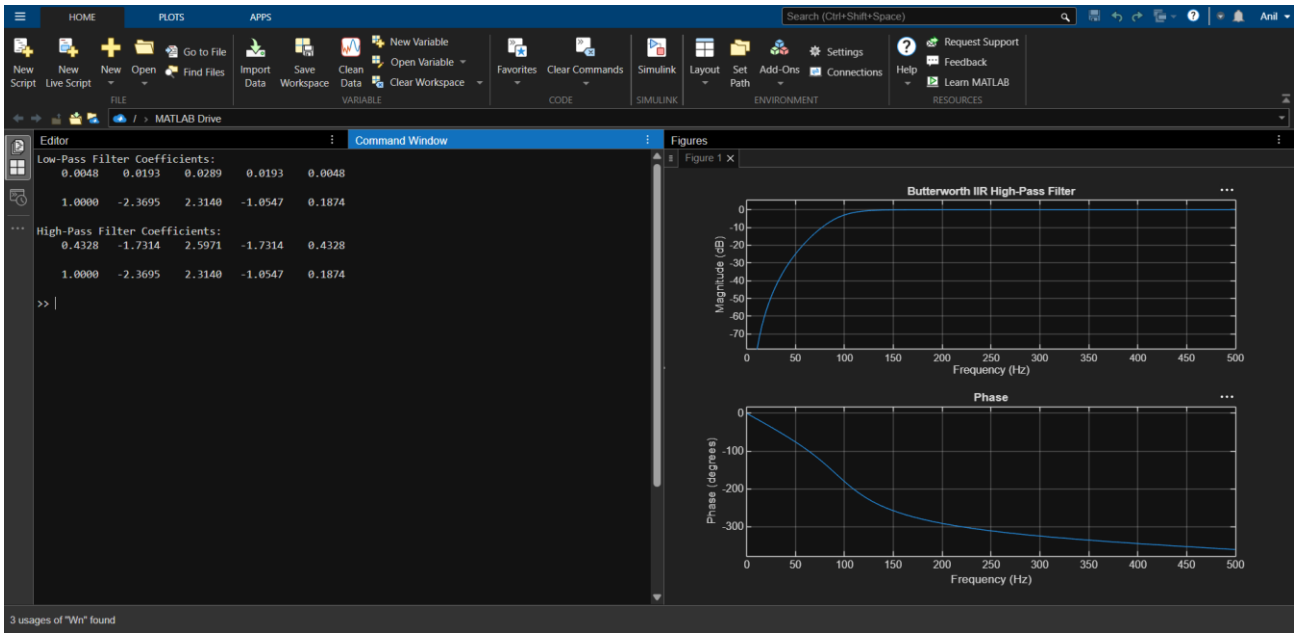
disp(a1);

disp('High-Pass Filter Coefficients:');

disp(b2);

disp(a2);
```

OUTPUT:



Objective 2:

To obtain and verify the filter coefficients and determine the required filter order.

CODE:

```
clc;

clear;

close all;

Fs = 1000;

Fp = 100;

Fst = 150;

Rp = 1;

Rs = 40;

Wp = Fp/(Fs/2);

Ws = Fst/(Fs/2);

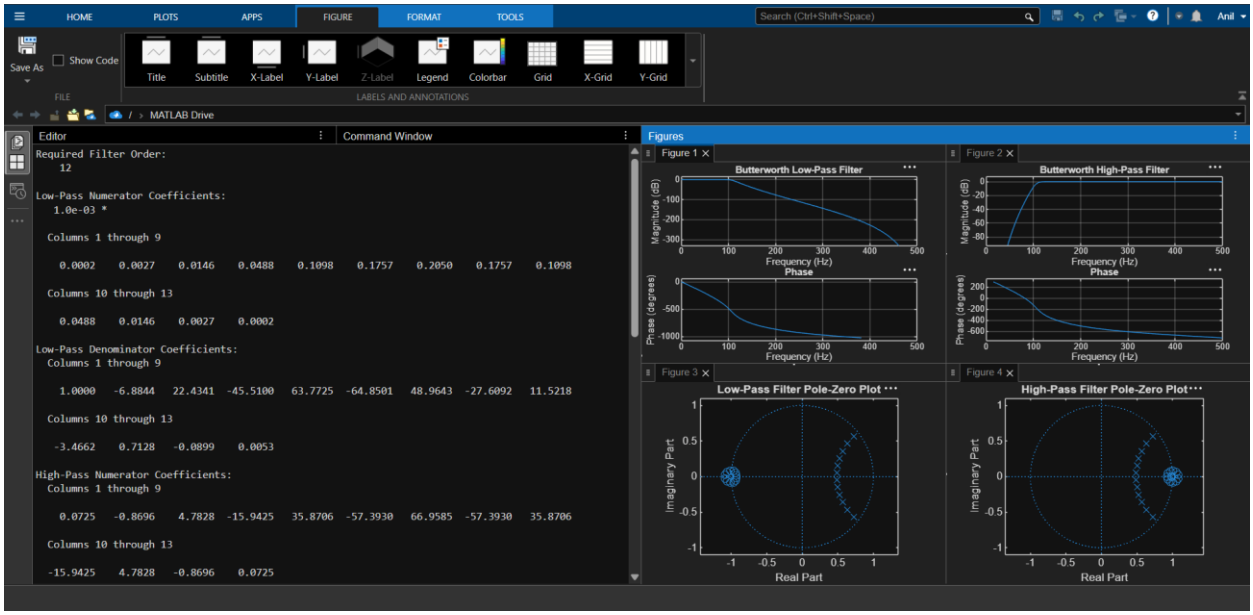
[N,Wn] = buttord(Wp,Ws,Rp,Rs);

[b_low,a_low] = butter(N,Wn,'low');

[b_high,a_high] = butter(N,Wn,'high');
```

```
disp('Required Filter Order:');
disp(N);
disp('Low-Pass Numerator Coefficients:');
disp(b_low);
disp('Low-Pass Denominator Coefficients:');
disp(a_low);
disp('High-Pass Numerator Coefficients:');
disp(b_high);
disp('High-Pass Denominator Coefficients:');
disp(a_high);
figure;
freqz(b_low,a_low,1024,Fs);
title('Butterworth Low-Pass Filter');
figure;
freqz(b_high,a_high,1024,Fs);
title('Butterworth High-Pass Filter');
figure;
zplane(b_low,a_low);
title('Low-Pass Filter Pole-Zero Plot');
figure;
zplane(b_high,a_high);
title('High-Pass Filter Pole-Zero Plot');
```

OUTPUT:



Objective 3:

To plot and analyze the magnitude and phase responses of the designed filters.

CODE:

```
clc;

clear;

close all;

Fs = 1000;

Fp = 100;

Fst = 150;

Rp = 1;

Rs = 40;

Wp = Fp/(Fs/2);

Ws = Fst/(Fs/2);

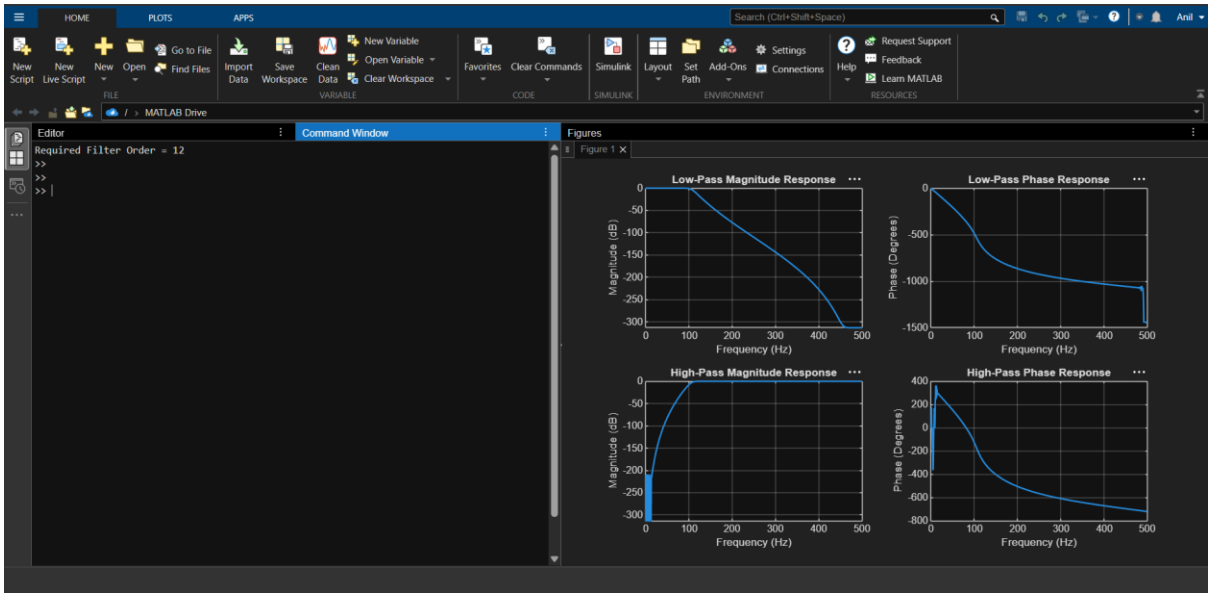
[N,Wn] = buttord(Wp,Ws,Rp,Rs);

[b1,a1] = butter(N,Wn,'low');
```

```
[b2,a2] = butter(N,Wn,'high');
[H1,f] = freqz(b1,a1,1024,Fs);
[H2,~] = freqz(b2,a2,1024,Fs);
figure;
subplot(2,2,1);
plot(f,20*log10(abs(H1)+eps),'LineWidth',1.5);
grid on;
title('Low-Pass Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
subplot(2,2,2);
plot(f,unwrap(angle(H1))*180/pi,'LineWidth',1.5);
grid on;
title('Low-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
subplot(2,2,3);
plot(f,20*log10(abs(H2)+eps),'LineWidth',1.5);
grid on;
title('High-Pass Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
subplot(2,2,4);
plot(f,unwrap(angle(H2))*180/pi,'LineWidth',1.5);
grid on;
title('High-Pass Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
```

disp(['Required Filter Order = ',num2str(N)]);

Output:



Objective 4:

To identify the cutoff frequency and examine the passband and stopband characteristics.

CODE:

```
clc;

clear;

close all;

Fs = 1000;

Fp = 100;

Fst = 150;

Rp = 1;

Rs = 40;

[Nlp,Wnlp] = buttord(Fp/(Fs/2),Fst/(Fs/2),Rp,Rs);

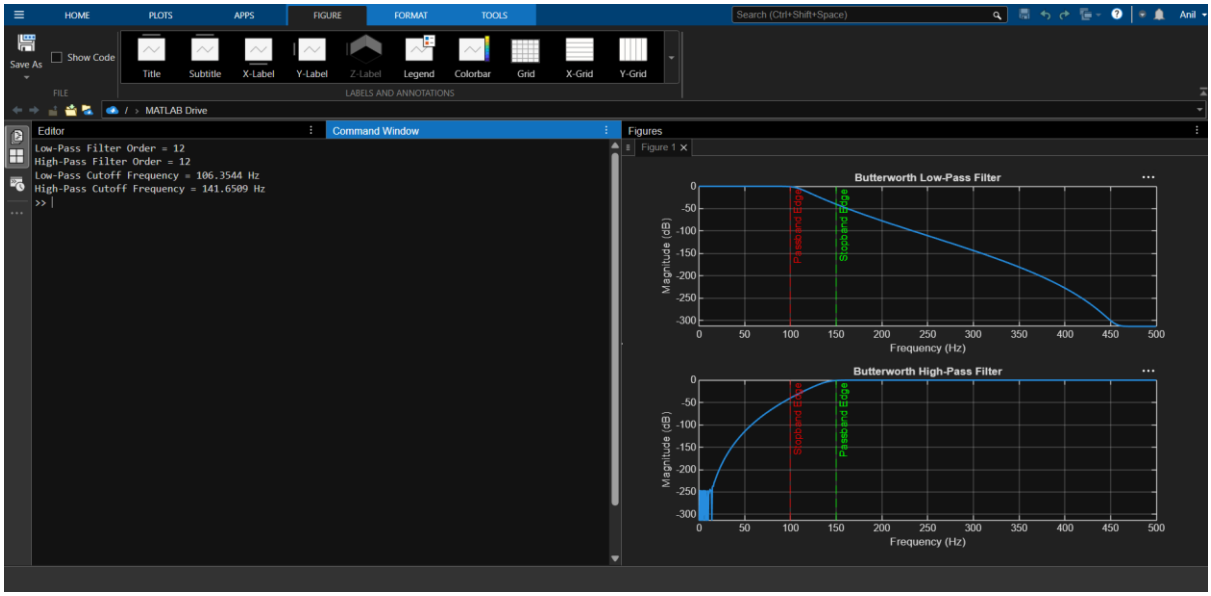
[blp,alp] = butter(Nlp,Wnlp,'low');

[Nhp,Wnhp] = buttord(Fst/(Fs/2),Fp/(Fs/2),Rp,Rs);

[bhp,ahp] = butter(Nhp,Wnhp,'high');
```

```
[Hlp,f] = freqz(blp,alp,1024,Fs);
[Hhp,~] = freqz(bhp,ahp,1024,Fs);
figure;
subplot(2,1,1);
plot(f,20*log10(abs(Hlp)+eps),'LineWidth',1.5);
grid on;
title('Butterworth Low-Pass Filter');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xline(Fp,'r--','Passband Edge');
xline(Fst,'g--','Stopband Edge');
subplot(2,1,2);
plot(f,20*log10(abs(Hhp)+eps),'LineWidth',1.5);
grid on;
title('Butterworth High-Pass Filter');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xline(Fp,'r--','Stopband Edge');
xline(Fst,'g--','Passband Edge');
disp(['Low-Pass Filter Order = ',num2str(Nlp)]);
disp(['High-Pass Filter Order = ',num2str(Nhp)]);
disp(['Low-Pass Cutoff Frequency = ',num2str(Wnlp*Fs/2),' Hz']);
disp(['High-Pass Cutoff Frequency = ',num2str(Wnhp*Fs/2),' Hz']);
```

OUTPUT


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Anil
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
Command Window
Figures
Figure 1
Low-Pass Filter Order = 12
High-Pass Filter Order = 12
Low-Pass Cutoff Frequency = 106.3544 Hz
High-Pass Cutoff Frequency = 141.6509 Hz
>>
Butterworth Low-Pass Filter
Passband Edge
Stopband Edge
Magnitude (dB)
Frequency (Hz)
Butterworth High-Pass Filter
Stopband Edge
Passband Edge
Magnitude (dB)
Frequency (Hz)